Voi / Portfolio Lab

Hub & Book investment evidence

Named report case | Model 7.2.0 | Frozen 2026-10-30 | Configuration cc19695f4f1a4859

The proposed priority is a Hub & Book pilot, followed by expansion only if customer response and delivery economics support the investment. This is a proposed strategic allocation of resources; the simulator continues to show competing designs and can recommend a different economic choice.

Financial measure	Proposed target	Improvement case	Reference case
Incremental discounted economic contribution	NOK 0.90m	+NOK 0.95m	+NOK 0.11m
Discounted economic payback	By month 24	Month 21	Month 33

The value target is a rounded management ambition, not a documented Voi hurdle. Payback by month 24 leaves 12 months within the appraisal horizon. The improvement case meets both targets under current inputs; the reference case does not meet them.

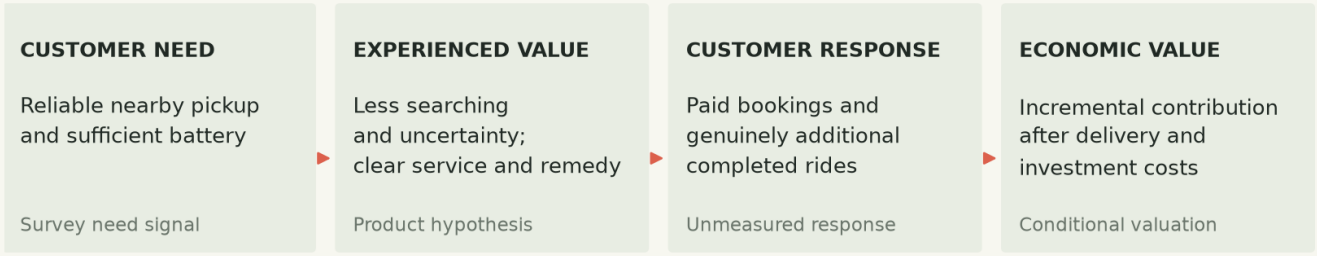
The improvement objective is 20% dedicated additional placement while maintaining 95% fulfilment. The scope is an estimated 124,822 active Oslo customers, phased expansion from month 10, and continued operating rights. These figures are for the full pilot-and-expansion programme, not a small pilot alone.

Pilot then scale scenario	Placement	Economic value	Payback	Peak cash
Improvement objective	20%	+NOK 0.95m	Month 21	NOK 623,353
Reference	25%	+NOK 0.11m	Month 33	NOK 663,342
Higher placement stress	30%	-NOK 0.85m	Not within 36 months	NOK 905,782

The reference uses 25% placement. At 30%, value is -NOK 0.85m. Only placement share changes between these cases; this is a sensitivity test, not a probability interval.

36-month horizon. Assumed annual discount 10%. No terminal value. Default relationship uplifts are zero. Resource costs and cash are distinguished; an unspecified funding ceiling does not confirm affordability.

Customer value and the operating argument



Customer need supports relevance. It does not establish demand for this exact booking product or a treatment effect.

Convenience Seekers rated adequate battery at 6.68/7 and nearby availability at 6.23/7 in the 72-person convenience sample. These descriptive need signals motivate a reliability proposition; they do not estimate booking adoption, population shares or willingness to pay.

Assumption	Planning value	Argument and evidence needed
Hub concentration	20% additional placement; 40% eligible reach	Select approved hubs and time windows with existing usable supply. Reduced placement must be achievable without simply losing customers. Measure geocoded requests, failed searches and incremental crew jobs.
Service reliability	95% fulfilment	Verify a usable, sufficiently charged vehicle at the promised location and time. Record failures and remedies; measure reliability jointly with placement and reach.
Price and response	NOK 35 fee; 8 bookings per adopter/month	A convenience need does not establish willingness to pay. Test fee conversion and repeat booking among eligible customers; include non-users in the economic evaluation.
Demand response	15% frequent / 30% subscriber adoption at the entered fee	Adoption is price-adjusted from a reference fee; both rates and the response coefficient remain assumptions. Record conversion at tested prices and compare assigned cohorts.
Scale and unit economics	124,822 estimated active customers; 23.06 net NOK/ride	Oslo scale and revenue allocation are constructed estimates. The survey is not representative. Validate city customer counts, segment reach and marginal costs before committing to scale.
Internal resource cost	100% of internal office-hour cost treated as new cash	Keep the full economic opportunity cost. Finance must distinguish hiring, contractor or replacement cash expenditure from existing salaried capacity.

The hypothesis must work jointly: choosing better-supplied hubs can lower placement cost, but a narrower footprint may also reduce customer relevance and demand. The model does not estimate that joint trade-off automatically.

A worked delivery unit

A dedicated placement costs NOK 73 under the entered 8 minutes, NOK 400/hour and NOK 20/job transport assumptions. At 20% placement, this is NOK 15 per booking attempt before overtime. Fee revenue is net of VAT and conditional on fulfilment; support, compensation, displacement and payment costs also apply.

Investment choices and uncertainty

Each investment faces the same customers, external demand path and no-offer counterfactual. The two implementation paths are shown separately; they must not be ranked on mixed bases.

Investment	Pilot value	Commercial value	Pilot peak cash	Pilot payback
No investment	NOK 0.00m	NOK 0.00m	NOK 0	-
Hub & Book	+NOK 0.95m	+NOK 1.41m	NOK 623,353	Month 21
Voi Together	-NOK 0.13m	-NOK 0.02m	NOK 375,000	Not within 36 months
Both current designs	+NOK 0.97m	+NOK 1.54m	NOK 930,050	Month 24

The commercial catalogue prefers Hub paid booking + coordination + split payment at +NOK 1.72m. Its pilot path gives +NOK 1.07m and needs NOK 986,902 at peak. Choosing Hub & Book first therefore also requires an argument about capital, delivery scope, customer value and evidence; this pack does not declare it the arithmetic winner.

Feasibility and funding

For the selected improvement path: Funding ceiling unspecified. Launch cash is NOK 409,733; peak cash is NOK 623,353. Launch-budget flags and the hard total-cash ceiling are separate checks. Neither positive value nor an affordable launch guarantees that the whole path can be financed.

Paired uncertainty on the pilot path

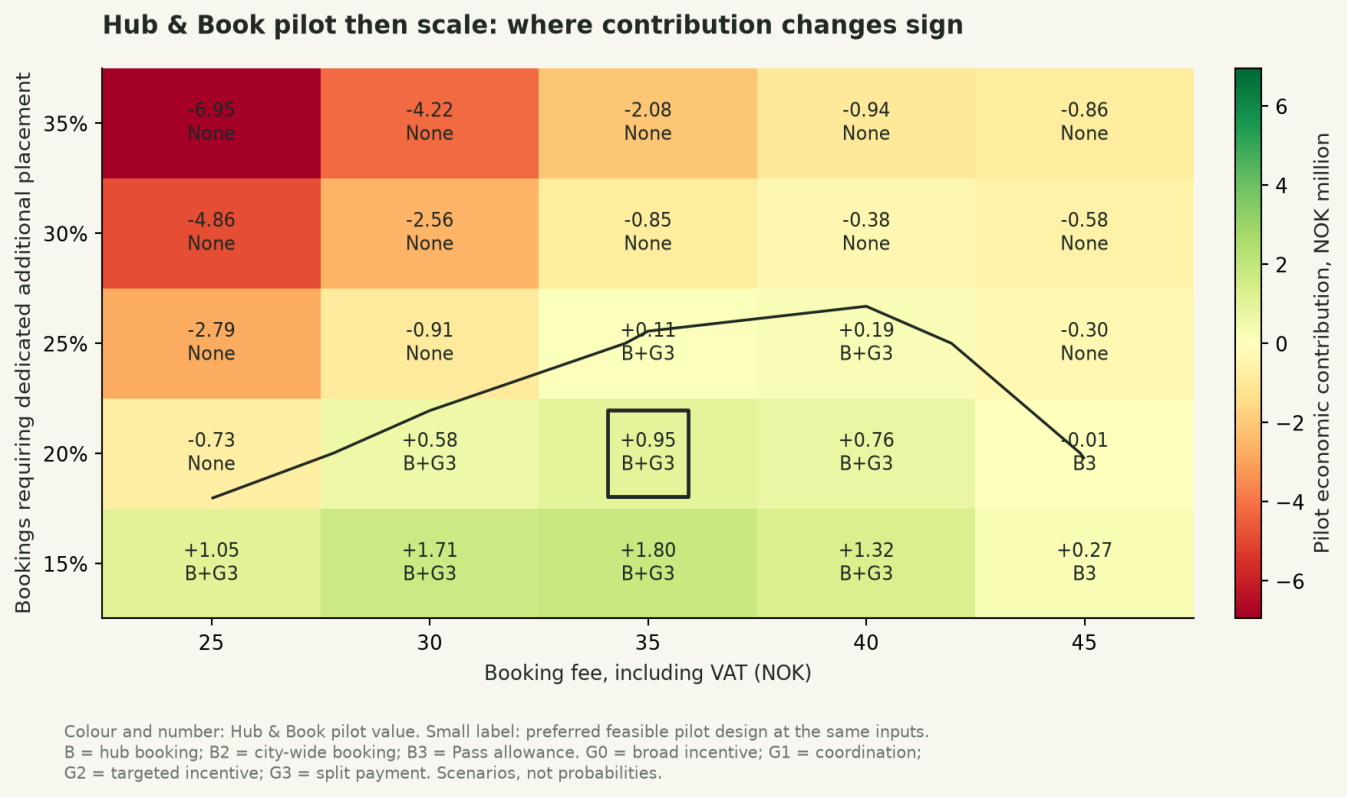
Candidate	5th percentile	Median	95th percentile	Below zero
No investment	NOK 0.00m	NOK 0.00m	NOK 0.00m	0.0%
Hub paid booking + coordination + split payment	-NOK 1.61m	+NOK 0.26m	+NOK 2.60m	41.6%
Hub & Book	-NOK 1.57m	+NOK 0.20m	+NOK 2.49m	43.0%
Voi Together	-NOK 0.47m	-NOK 0.23m	+NOK 0.23m	82.0%
Both current designs	-NOK 1.69m	+NOK 0.15m	+NOK 2.48m	44.6%

3,000 paired draws; seed 2026. The selected booking option and no investment are included. Values follow the pilot path. Shares describe assumed combinations, not measured chances of success. Range symmetry and dependence are modelling choices; changed ranges are not improved product evidence.

The target does not price unknown unknowns

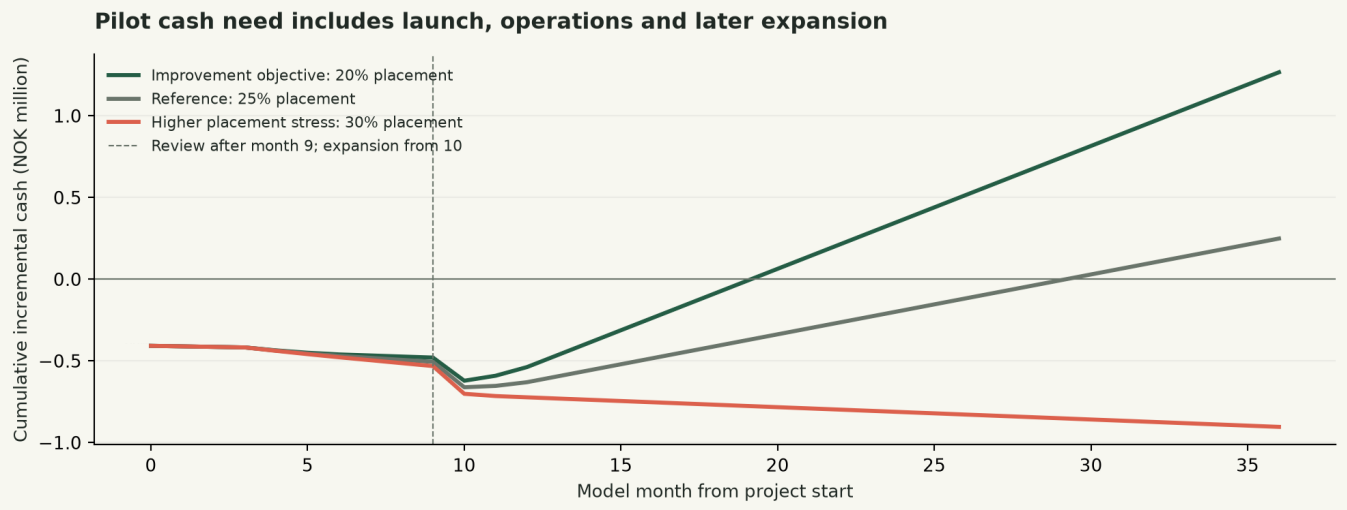
Reassess the case when operating permissions, rival pricing, technology or customer behaviour changes materially. Switching costs, satisfaction and actual repeat use are distinct. Current economics does not establish durable protection from imitation.

Decision boundary and cash



Colour and number: Hub & Book pilot value. Small label: preferred feasible pilot design at the same inputs.
B = hub booking; B2 = city-wide booking; B3 = Pass allowance. G0 = broad incentive; G1 = coordination;
G2 = targeted incentive; G3 = split payment. Scenarios, not probabilities.

Values and the zero line follow the selected Hub & Book pilot path. Small labels identify the preferred feasible pilot design at the same fee and placement assumptions. The report choice and the highest-value catalogue design remain distinct.



The improvement, reference and stress cash curves come from their frozen monthly rows. Economic value retains full internal resource cost. Additional internal cash share is 100%; a different staffing plan changes cash needs without removing the opportunity cost.

Pilot delivery and decision protocol

Milestone	Model timing	Interpretation
First launch	Month 4	Benefits start after required capabilities are ready.
First cohort completes ramp	Month 7	Later expansion cohorts each have their own ramp.
Operational review	After month 9	Expansion starts in month 10 only if the review supports it.
Retention evidence window	Ends month 13	Six complete post-ramp months; any claim needs concurrent control through its window.
Control release	After month 9	15% of the market is held as control until release.

Economic outcome and customer guardrails

Measure incremental contribution per assigned eligible customer, including non-bookers, against a concurrent control. Use stable hub-zone assignment, record spillover and analyse at the assignment level. Pre-specify fulfilment, battery sufficiency, customer effort, complaints and refund guardrails. Measure fee conversion, repeat booking, placement minutes and displaced trips together.

Track the full-programme appraisal against NOK 0.90m and payback by month 24. Approve expansion only if future incremental contribution exceeds remaining avoidable costs, service reliability is maintained, and cash/capacity constraints are satisfied. Exclude sunk setup from that continuation decision; any proposed target change returns to the board. The month-9 gate does not validate a retention effect; no such uplift is credited here.

Owner	Accountability
Country manager / programme owner	Sponsor, cross-functional authority, review and escalation.
Operations	Approved hub locations; qualified supply; capacity, crew jobs and fulfilment records.
Product and Commercial	Service promise, booking flow, price tests, terms and remedies.
Analytics	Assignment, contamination, assigned-customer outcomes and updated appraisal.
Finance	Economic cost versus additional cash, funding, and continuation assessment.

Study precision is not assumed

Required inputs include eligible participants, independent clusters, cluster size, ICC, outcome variance, attrition and minimum worthwhile effect. Until these are supplied, there is no sample-size or detectable-effect claim. Few clusters and interference need specialist design validation.

Three genuine first-time readers must still complete the five-minute protocol. Automated browser and numerical checks do not establish usability or empirical validity.

Methods, report use and references

Required report section	Use of this evidence
Executive Summary	State Hub & Book first, pilot then conditional expansion; identify the principal economic dependency.
Situation Analysis	Use the needs-based segment and reliability gap; distinguish survey description from market estimates.
Financial Goals	Propose at least NOK 0.90m economic contribution and discounted payback by month 24; compare against the frozen scenarios.
Strategic Initiatives and Priorities	Specify the hub service, eligibility, price, operational promise and why Together is deferred or retained as an alternative.
Budget	Separate economic resource cost, actual cash launch needs, later capacity and peak cash. Funding remains conditional on an entered ceiling.
Implementation and Control	Assign owners; measure contribution per assigned customer and reliability; review at the gate, and refresh the investment appraisal.

Methods and limits

Economic contribution is the discounted difference between each offer and the same no-offer world, less resource setup. All configurations use 36 months and 10% annual discount. Monthly cash instead uses incremental cash expenditure. The active-customer scale, Oslo revenue allocation, adoption, additionality and operating costs remain assumptions or derived proxies. Usage states do not measure relationship quality.

The public data audit did not establish a compatible historical Voi scooter-trip series. Prophet remains off. Imported demand factors must affect the offer and counterfactual consistently. A forecasting model would require historical validation against a seasonal baseline.

Complete source and assumption register, monthly results, methods and pilot protocol accompany this PDF. Configuration cc19695f4f1a48599f5bccc5995f3d76110c861d4ca54f517ae45911868e9331. Figures and tables use the same frozen inputs. The core report must remain understandable without opening this supplementary pack.

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